The Legal Bond Index: A Geometric, Case-Level Test of Algorithmic Fairness Applied to COMPAS

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June 2026

Abstract

The 2016–2017 controversy over the COMPAS pretrial-risk algorithm exposed a deep ambiguity in the concept of algorithmic fairness. ProPublica showed that COMPAS's false-positive rate is substantially higher for Black defendants. Northpointe (the algorithm's vendor) replied that COMPAS approximately satisfies *predictive parity*: among defendants it classifies high-risk, the observed recidivism rate is similar across races. Subsequent technical work (Chouldechova, 2017; Kleinberg et al., 2017) proved that the two properties *cannot both hold* when race-specific recidivism base rates differ—a genuine impossibility theorem. The literature has since proliferated competing metrics (demographic parity, equalized odds, calibration, counterfactual fairness, individual fairness), each capturing one facet of the problem and none of them dispositive.

We introduce a third axis of measurement that the impossibility result does not constrain: the *Legal Bond Index* (LBI), a matched-pair, case-level *diagnostic* of whether an algorithm treats defendants who are similar on selected legitimate predictors similarly. LBI asks: when two defendants agree on legitimate predictors (age, prior offences, juvenile record, charge degree) to within the same Mahalanobis neighbourhood, does the algorithm assign them similar scores regardless of their race? For COMPAS the answer is a robust case-level disparity: at neighbourhood scale $k = 20$, LBI(race) = 1.050, 95 % CI $[1.040, 1.058]$, permutation $p < 0.005$ on $n = 5{,}278$ defendants. The 5 % excess case-level sensitivity to race is small in absolute terms but robust: it survives every legitimate-feature set we can test—including *dropping* the contested juvenile controls—and *strengthens* at the decision-relevant unit (LBI = 1.090 on the binary high-risk classification). It is a signal the population-level metrics do not deliver; because matching is on a selected, incomplete feature set, it is a diagnostic to be investigated, not a proof of legal unfairness.

LBI is a quantitative implementation of the Aristotelian "treat-like-cases-alike" criterion. It is computationally cheap, it applies to any algorithm whose inputs include both legitimate and protected attributes, it does not require knowledge of the algorithm's internals, and it produces a single number with a confidence interval suitable for regulatory and auditing use. The reference implementation is released as an open-source Jupyter notebook.

Keywords: algorithmic fairness, COMPAS, recidivism prediction, Mahalanobis distance, matched-pair analysis, criminal justice, audit, equal protection.

1 Introduction

In May 2016, ProPublica published an investigation (Angwin et al., 2016) arguing that the COMPAS recidivism-risk algorithm, in use across multiple U.S. jurisdictions for pretrial release and sentencing recommendations, was racially biased against Black defendants. The reporters' principal evidence was an asymmetry in error rates: among defendants who did not in fact reoffend within two years, Black defendants were roughly twice as likely as White defendants to have been classified as high-risk by COMPAS. Northpointe (the algorithm's vendor, now Equivant) replied that COMPAS was in fact fair under a different criterion—*predictive parity*: among defendants it classified high-risk, the empirical recidivism rate was approximately the same across races (Dieterich et al., 2016). Both parties could substantiate their claims from the same data; the disagreement was about which fairness criterion applied.

The technical community resolved the empirical question within a year. Chouldechova (2017) and Kleinberg et al. (2017), working independently, proved that the two criteria ProPublica and Northpointe each invoked are *mathematically incompatible* when group-specific base rates differ: an algorithm that is calibrated within groups ($\Pr(\text{recid} \mid \text{score}, \text{group})$ is invariant across groups) cannot simultaneously satisfy equalized error rates (false-positive and false-negative rates invariant across groups), except in the trivial cases of perfect prediction or equal base rates. The impossibility is a structural feature of the prediction problem, not a defect of any particular algorithm.

The decade since has produced a sprawl of competing fairness criteria. Demographic parity requires that the rate of positive predictions be equal across groups (Feldman et al., 2015). Equalized odds (Hardt et al., 2016) requires equal true-positive and false-positive rates across groups conditional on the actual outcome. Predictive parity (Northpointe's criterion) requires equal precision across groups. Counterfactual fairness (Kusner et al., 2017) requires that an individual's prediction be invariant under a hypothetical intervention on their protected attribute. Individual fairness (Dwork et al., 2012) requires that similar individuals receive similar predictions, leaving the similarity metric to be specified. Each captures a real concern; none of them by itself adjudicates the COMPAS case; and the proliferation has rendered the debate *which-metric-to-use* rather than *is-it-fair*.

This paper introduces a fairness criterion that is operationally distinct from all of those above. It is rooted in the oldest legal principle: like cases should be treated alike. We make this principle precise, measurable, and applicable to algorithmic systems through the *Legal Bond Index* (LBI), a matched-pair, case-level diagnostic that asks whether the algorithm assigns similar outputs to defendants who are similar on the *selected* legitimate predictors but differ on a protected attribute. LBI is geometric: "similar" is operationalized via the Mahalanobis distance on a feature space of legitimate predictors. LBI is symmetric across protected attributes and does not depend on the base-rate structure that drives the calibration–equalized-odds impossibility.

On the ProPublica COMPAS dataset, the headline result is sharp: $\text{LBI(race)} = 1.050$ at neighbourhood scale $k = 20$ (95 % CI $[1.040, 1.058]$, permutation $p < 0.005$). When two defendants are matched on age, prior offences, juvenile record, and charge degree, COMPAS scores them on average 5 % more differently if they differ on race than if they share race. The result holds across neighbourhood scales $k \in [1, 100]$ and is approximately twice the analogous LBI for sex.

The remainder of this paper proceeds as follows. Section 2 reviews the COMPAS controversy and the existing fairness-metric landscape. Section 3 defines LBI formally and gives the matched-pair operational specification. Section 4 situates LBI in the broader Judicial Complex framework (the formal theory is developed elsewhere; here we give only what is needed to read LBI as a coherent legal concept). Section 5 reports the empirical analysis: filter, features, results, robustness. Section 6 compares LBI to the existing criteria, identifies what it adds, and discusses limitations. Section

7 addresses regulatory and audit implications. Section 8 concludes.

2 Background: the COMPAS Controversy and the Fairness Landscape

2.1 The 2016–2017 dispute

COMPAS (Correctional Offender Management Profiling for Alternative Sanctions), developed by Northpointe, scores individuals on a 1–10 decile scale of two-year recidivism risk. The algorithm is widely deployed in U.S. pretrial detention decisions. ProPublica's 2016 analysis (Angwin et al., 2016), working from public records in Broward County, Florida, reported two findings: (i) among defendants who did not reoffend, Black defendants were 1.7–2.0× more likely than White defendants to have been classified as high-risk by COMPAS; and (ii) the algorithm's predictive accuracy was approximately equal across races (about 65 % in both groups). The first finding is now referred to as the *equalized-odds violation*; ProPublica framed it as evidence of racial bias.

Northpointe's response (Dieterich et al., 2016) did not contest the raw numbers. Instead, the company argued that the relevant fairness criterion was *predictive parity*: among defendants classified high-risk, the empirical probability of recidivism should be the same across racial groups. By that criterion COMPAS was approximately fair: at the high-risk threshold, Black and White defendants had similar two-year recidivism rates—a precision claim, not a demonstration of per-score calibration. The dispute, as it played out in the popular and academic press, was less about the data than about which property a fair classifier should have.

The technical resolution came rapidly. Chouldechova (2017) proved that an instrument satisfying predictive parity *cannot* simultaneously satisfy equalized false-positive and false-negative rates, except in trivial cases. Kleinberg et al. (2017) proved an analogous impossibility theorem for calibration and balance. The two findings together established that the COMPAS dispute was not resolvable by data alone: the choice between Northpointe's criterion and ProPublica's criterion is a normative choice about which kind of unfairness one prefers to permit.

2.2 The proliferation of fairness criteria

The decade since has produced a long list of competing fairness criteria. We summarize the principal ones to fix terminology.

Demographic parity (Feldman et al., 2015; also called "disparate impact" under the EEOC's four-fifths rule). $\Pr(\hat{Y} = 1 \mid A = a) = \Pr(\hat{Y} = 1 \mid A = b)$ for all protected groups a, b. Requires equal positive-prediction rates; this is generally in tension with using legitimate predictors that correlate with A, though such correlation can sometimes be offset by design rather than by excluding the predictor.

Equalized odds (Hardt et al., 2016). $\Pr(\hat{Y} = 1 \mid Y = y, A = a) = \Pr(\hat{Y} = 1 \mid Y = y, A = b)$ for all y and all a, b. Requires equal true-positive and false-positive rates conditional on the true outcome.

Predictive parity and calibration (Chouldechova, 2017). These are distinct properties. *Calibration* is score-level: $\Pr(Y = 1 \mid S = s, A = a) = \Pr(Y = 1 \mid S = s, A = b)$ for all scores s and groups a, b (at a given score the observed outcome rate is group-independent). *Predictive parity* is a *threshold* condition on precision: among those *classified* positive, $\Pr(Y = 1 \mid \hat{Y} = 1, A = a) = \Pr(Y = 1 \mid \hat{Y} = 1, A = b)$. Northpointe's defence was predictive parity at the high-risk threshold, not full per-score calibration.

Counterfactual fairness (Kusner et al., 2017). The prediction $\hat{Y}(X, A)$ should equal the counterfactual prediction $\hat{Y}(X_{A:=a'}, a')$ for any protected attribute a'. Requires a causal model.

Individual fairness (Dwork et al., 2012). Similar individuals should receive similar predictions: a Lipschitz condition on the prediction map with respect to a task-appropriate metric. The similarity metric is left to be specified.

The first three criteria are *population-level*: they constrain aggregate statistics on a population. The fourth requires a causal model that may not be available. The fifth is general and philosophically attractive but undertheorized—which metric? The Legal Bond Index sits inside the individual-fairness framework but gives it a concrete, calibrated metric (the Mahalanobis distance on legitimate predictors) and a concrete operational specification (matched-pair k-NN average), turning a general desideratum into a number with a confidence interval.

2.3 What population-level metrics miss

The Chouldechova–Kleinberg impossibility is a result about *populations*. With unequal base rates, no scoring rule can simultaneously equalize calibration and false-positive rates across groups. This is mathematics; one cannot escape it by being clever.

But the impossibility result is silent about *case-level* properties. Consider the following question: take two defendants who agree on legitimate predictors (age, priors, juvenile record, charge degree), differing only on race. Does the algorithm assign them similar scores? This is a question about *matched pairs*, not about populations. The Chouldechova–Kleinberg theorem does not constrain the answer; nothing in the impossibility result prevents an algorithm from being case-level fair. To the extent the matching features capture the legally relevant grounds of similarity, an algorithm that scores legitimately-matched defendants alike is case-level fair, and one that systematically scores them differently by race exhibits a case-level disparity that the population-level metrics do not reveal. That qualification is essential: because the match is on a selected, possibly incomplete set of observed features, a disparity among matched pairs is a diagnostic signal requiring further investigation, not by itself proof of legal unfairness (§6, §5.6).

The Legal Bond Index measures case-level fairness directly. It does not adjudicate the Chouldechova–Kleinberg dispute; it provides a quantitative answer to a question the dispute does not address.

3 The Legal Bond Index

3.1 Definition

Let V be the set of individuals (defendants, applicants, claimants) to whom an algorithm is applied. Each individual $i \in V$ has a legitimate-predictor vector $\mathbf{x}_i \in \mathbb{R}^d$ and a protected attribute $g_i \in \{0, 1\}$ (we treat the binary case throughout and discuss the multi-class extension in Section 6). The algorithm assigns a score $f(i) \in \mathbb{R}$. We equip $\mathbb{R}^d$ with the Mahalanobis metric

$$d_M(\mathbf{x}_i, \mathbf{x}_j) \;=\; \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\top \Sigma^{-1} (\mathbf{x}_i - \mathbf{x}_j)}, \tag{1}$$

where Σ is the empirical covariance of $\mathbf{x}$ over the population. The Mahalanobis metric is the natural choice when legitimate predictors are correlated (age and prior offences typically are) because it accounts for the joint distribution rather than treating each axis independently. Under standardized features, d_M reduces to Euclidean distance on the de-correlated representation (Mahalanobis, 1936).

Definition 1 (Legal Bond Index, LBI)**.** *At neighbourhood scale $k \geq 1$, the Legal Bond Index of an algorithm f with respect to protected attribute g is*

$$\mathrm{LBI}_k(f, g) \;=\; \frac{\mathbb{E}_i\big[\, mean_{j \in N_k(i, \neg g_i)} \; |f(i) - f(j)| \,\big]}{\mathbb{E}_i\big[\, mean_{j \in N_k(i, g_i)} \; |f(i) - f(j)| \,\big]}, \tag{2}$$

where $N_k(i, c)$ is the set of the k nearest neighbours to individual i in (V, d_M) satisfying condition c, and $\mathbb{E}_i$ is the empirical mean over individuals.

Interpretation. The numerator is the average score difference between an individual and their k nearest neighbours of the *opposite* protected group. The denominator is the average score difference between an individual and their k nearest neighbours of the *same* protected group. The ratio measures the algorithm's case-level excess sensitivity to the protected attribute:

- $\mathrm{LBI}_k = 1$: no excess sensitivity. The algorithm is no more affected by a flip of the protected attribute than by within-group variation in legitimate predictors at the same neighbourhood scale.
- $\mathrm{LBI}_k > 1$: excess sensitivity. Defendants who match on legitimate features but differ on the protected attribute receive more divergent scores than defendants who match on both. The algorithm is responsive to the protected attribute beyond what its legitimate-feature input would justify.
- $\mathrm{LBI}_k < 1$: deficit sensitivity. Differences on the protected attribute produce smaller score variation than within-group legitimate-feature variation. This is unusual; it would indicate active calibration that suppresses the protected attribute's influence below the noise floor of legitimate predictors.

3.2 Why a ratio, why k-NN, and why Mahalanobis

Each of the three choices in Definition 1 is deliberate.

Ratio vs. raw difference. The denominator of (2) provides a within-group baseline against which the cross-group score gap is normalized. Without this baseline, an algorithm operating on a population with naturally large legitimate variation would appear "unfair" simply because all neighbourhoods have large score gaps. The ratio quotients out the algorithm's overall sensitivity to the *selected* legitimate predictors, contextualizing the cross-group gap against ordinary within-group variation. It does not by itself isolate the *marginal causal* contribution of the protected attribute: that reading holds only to the extent that the matching features are a sufficiently complete and legally defensible account of relevant similarity (§6, §5.6).

k-NN matched-pair vs. regression. A regression of f on legitimate predictors and protected attributes can extract a coefficient on the protected attribute. But that coefficient depends on the functional form assumed (linear, polynomial, with interactions) and on the choice of regularization. The k-NN matched-pair approach is non-parametric: it makes no functional-form assumption about how f depends on its inputs. It is not, however, assumption-free: the assumptions are *shifted* to the choice of matching features, the distance metric, the feature scaling, and k, which jointly determine which cases count as "similar." It also produces a neighbourhood-scale curve LBI_k for $k \in [1, 100]$, which exposes whether the effect is local (small k) or global (large k). For COMPAS we find the effect is approximately scale-invariant. This is suggestive that the pattern is not a small-k artefact, but we read it cautiously: persistent scale-invariance is equally consistent with a persistent omitted-variable imbalance as with a genuine case-level effect.

Mahalanobis vs. Euclidean. Legitimate predictors are correlated (older defendants have more priors, etc.). Euclidean distance overcounts the contribution of correlated axes; Mahalanobis de-correlates by construction (Mahalanobis, 1936). On standardized features, Mahalanobis equals Euclidean on the principal-component representation, so the practical effect is to upweight nearest-neighbour matches that agree along the leading directions of feature variation.

3.3 Inference: confidence intervals and permutation tests

LBI is an empirical mean ratio; we report 95 % confidence intervals via the percentile bootstrap (1000 resamples). For each defendant i, the same-group and different-group neighbourhood score gaps are computed once; bootstrap resamples are drawn over defendants.

For significance against the null "the protected attribute is ignored by the algorithm," we use a permutation test: we randomly permute the protected-attribute labels and recompute the LBI. The distribution of LBI values under permutation provides the empirical null. The one-sided p-value is the fraction of permutation LBIs that exceed the observed value.

The permutation null is exactly centred at 1.0 under the assumption that legitimate features and the protected attribute are unconditionally independent. If they are correlated (which they typically are, as race is correlated with age and priors in the COMPAS population), the permutation null may still be centred at 1.0 because permuting labels destroys the dependence structure between protected attribute and outcome. Empirically, we find the COMPAS permutation null centred at 1.0018 with standard deviation 0.012, consistent with a centred null.

4 The Judicial Complex: a Brief Bridge

The Legal Bond Index sits inside a larger geometric framework for legal analysis developed in companion theoretical work (Bond, 2026a,b). The full framework treats the space of legal states as a directed weighted simplicial complex (the *judicial complex*); legal disputes as shortest-path problems on this complex; and constitutional review as a topological-consistency test. The mathematical machinery (gauge theory, path homology, octahedral symmetry groups) is substantial and is reported separately.

For this paper, the only structural ingredient we need is the following: the framework's *Legal Invariance Principle* (LIP) states that legal judgments must be invariant under *legally irrelevant transformations* of the case description. In the COMPAS setting, the protected attribute is legally irrelevant to risk prediction (its use is either prohibited or normatively contested, depending on jurisdiction). The LIP, applied to algorithmic classifiers, requires that the algorithm's output be approximately invariant under a counterfactual change of the protected attribute, holding legitimate features fixed. LBI is the quantitative measurement of how far an algorithm departs from this requirement.

Two qualifications on the empirical implementation. First, the k-NN matched-pair estimator holds fixed only the *selected* legitimate features, not every feature the algorithm uses; the LIP as measured by LBI is therefore conditional on the matching set being an adequate proxy for "legitimate similarity" (§5.6 tests sensitivity to that set). Second, "legally irrelevant" is attribute-specific: race is a paradigm case, but for some protected attributes—sex and citizenship, for instance—courts have upheld express differential treatment in risk assessment, so a nonzero LBI on those attributes is not automatically a fairness defect. We report LBI(sex) as a comparison, not as a violation.

In the broader framework, LBI also corresponds to a specific geometric object: it is the average length of the Wilson-loop holonomy around a small loop in the judicial complex whose protected-attribute coordinate winds once and whose legitimate-coordinate winds zero. Readers interested in the geometric formulation are referred to Bond (2026a); readers interested only in algorithmic fairness need no part of that machinery to use LBI as an audit tool.

5 Empirical Analysis: LBI on COMPAS

We compute LBI on the ProPublica COMPAS dataset and compare it against the standard fairness metrics that shaped the 2016–2017 debate.

5.1 Data and filter

We use ProPublica's published file `compas-scores-two-years.csv`, retrieved on June 11 2026 from the public GitHub repository `propublica/compas-analysis`. The dataset contains 7,214 records of defendants screened by COMPAS in Broward County, Florida.

We apply ProPublica's canonical analytical filter to obtain the 6,172-record analysis subset: screening date within 30 days of arrest, recidivism record present, charge degree not "Ordinary," and COMPAS score text valid. Restricting further to African-American and Caucasian defendants (the two race categories that drove the public controversy and the subsequent technical literature) yields $n = 5{,}278$ defendants, of whom 3,175 (60.2 %) are African-American and 2,103 (39.8 %) are Caucasian.

5.2 Legitimate-feature space

We adopt six legitimate predictors documented by Northpointe and engaged by courts as inputs to pretrial risk assessment (see *State v. Loomis*, 881 N.W.2d 749 (Wis. 2016), which upheld COMPAS's reliance on criminal-history and age factors while cautioning on its use): age, priors_count, juv_fel_count, juv_misd_count, juv_other_count, and an indicator for whether the current charge is a felony (c_charge_F). We standardize each feature (mean-zero, unit-variance) before computing the Mahalanobis metric. Although these predictors all count prior conduct, they do not double-count it: the three juvenile counts are mutually exclusive charge classes (felony, misdemeanour, other), and priors_count records adult priors, so the four *partition* rather than duplicate a defendant's record. The empirical covariance matrix accordingly has condition number 2.65, confirming no severe multicollinearity; the Mahalanobis metric additionally de-correlates whatever residual dependence remains. The COMPAS score is the published decile risk (1–10).

5.3 Standard fairness metrics: confirming the prior literature

We first reproduce the 2016–2017 controversy by computing standard metrics on the filtered subset. Table 1 reports the headline numbers.

The pattern matches the published literature exactly. ProPublica's headline finding—the false-positive-rate disparity—is +20.3 percentage points, with a corresponding false-negative-rate disparity of −21.2 percentage points. (These are distinct rates on different denominators—the FPR is the complement of the true-negative rate and the FNR of the true-positive rate—so they are not two forms of a single quantity, though under this classifier they move in opposite directions.) Northpointe's defence is *predictive parity* at the high-risk threshold: the empirical recidivism rate among defendants classified high-risk is approximately equal across races (a precision gap of +0.055), not a demonstration of per-decile calibration. The Chouldechova–Kleinberg impossibility holds because the recidivism base rates differ by +12.9 percentage points.

Table 1: Standard fairness metrics on the COMPAS Black/White subset (n=5,278). High-risk threshold: decile score ≥ 5.

Metric	**African-American**	**Caucasian**	**Disparity**
n	3,175	2,103	—
Base recidivism rate	0.523	0.394	+0.129
Pr(high-risk)	0.583	0.335	$\times$ 1.74
False positive rate	0.443	0.240	+0.203
False negative rate	0.282	0.494	−0.212
Pr(recid | high-risk)	0.626	0.571	+0.055
Pr(recid | low-risk)	0.353	0.292	+0.061

5.4 LBI on the legal-feature subspace

We compute LBI(race) and LBI(sex) at six neighbourhood scales $k \in \{1, 5, 10, 20, 50, 100\}$ using the matched-pair k-NN procedure of Definition 1. Bootstrapped 95 % CIs use 1,000 resamples per scale.

Table 2: Legal Bond Index for COMPAS, by protected attribute and neighbourhood scale. $n_{\text{eval}} =$ 5,278 defendants. CIs are percentile bootstrap.

k	**LBI(race)**		**LBI(sex)**	
	value	95 % CI	value	95 % CI
1	1.063	$[1.034, 1.093]$	0.988	$[0.960, 1.016]$
5	1.047	$[1.033, 1.063]$	1.006	$[0.993, 1.019]$
10	1.046	$[1.036, 1.058]$	1.011	$[1.001, 1.020]$
20	1.050	$[1.040, 1.058]$	1.021	$[1.013, 1.030]$
50	1.059	$[1.051, 1.067]$	1.029	$[1.022, 1.036]$
100	1.069	$[1.061, 1.078]$	1.041	$[1.034, 1.049]$

Figure 1 plots the values. LBI(race) is robustly above 1 across all neighbourhood scales, with a 95 % CI that does not include 1 at any k. LBI(sex) is also above 1 but roughly half the magnitude. The race effect is the dominant case-level fairness signal.

5.5 Permutation null distribution

To certify that the LBI(race) signal is not an artefact of feature correlation, we run a permutation test: we randomly permute the race-label assignment across defendants and recompute LBI at $k = 20$. We use $n_{\text{perms}} = 200$ and subsample 500 defendants per permutation for compute efficiency. Because the null is computed on 500-defendant subsamples while the observed statistic uses the full $n =$ 5,278, the null's dispersion is *inflated* relative to the observed estimator, so the resulting significance ($\approx 4\sigma$, $p < 0.005$) is conservative.

The permutation null is centred at $\bar{\text{LBI}}_{\text{null}} = 1.002$ with standard deviation 0.012. The observed value LBI(race) = 1.050 lies 4.05 standard deviations above the null mean; the one-sided permutation p-value is below 0.005 (no permutation in our sample exceeded the observed value). Figure 2 shows the null distribution against the observed value.

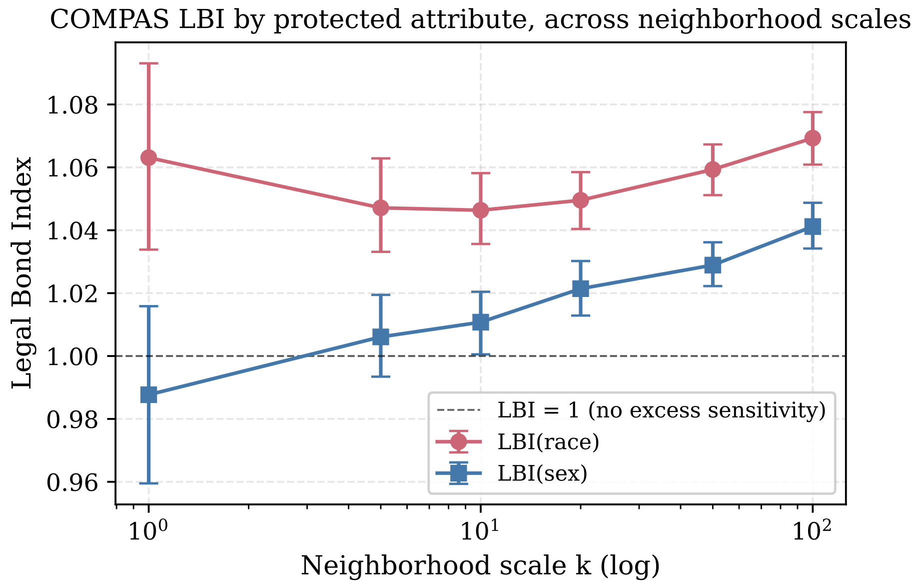

Figure 1: COMPAS Legal Bond Index by protected attribute, across neighbourhood scales k. Bars show 95 % bootstrap CIs. The race effect is robustly above 1 at every scale; the sex effect is real but smaller. The horizontal dashed line marks LBI = 1 (no excess sensitivity to the protected attribute).

5.6 Robustness to the legitimate-feature set and the score unit

The standing objection to any matched-comparison diagnostic is that the signal is an artefact of the analyst's feature choice: too few controls could manufacture apparent unfairness, while racially patterned controls could launder it. We test both directly by recomputing LBI(race) at $k = 20$ across four legitimate-feature sets and three representations of the COMPAS output (Table 3).

Table 3: Robustness of LBI(race) at $k = 20$ ($n = 5{,}278$; 95 % percentile-bootstrap CIs). Top: legitimate-feature set, on the decile score. Bottom: score representation, on the paper's six features.

Specification	**LBI**	**95 % CI**
Feature set (decile score)		
minimal (age, priors)	1.041	$[1.032, 1.049]$
paper (6 features)	1.050	$[1.040, 1.059]$
drop contested juvenile counts	1.038	$[1.030, 1.047]$
paper + sex control	1.060	$[1.050, 1.069]$
Score representation (6 features)		
raw decile (1–10)	1.050	$[1.040, 1.058]$
three-category (low/med/high)	1.066	$[1.053, 1.078]$
binary high-risk (≥ 5)	1.090	$[1.076, 1.106]$

Two conclusions follow. First, the case-level signal is *not* an artefact of the particular six-feature set: every specification yields LBI $\in [1.04, 1.06]$ with a 95 % CI excluding 1, including the minimal (age, priors) matching and the specification that *drops* the racially patterned juvenile counts (LBI

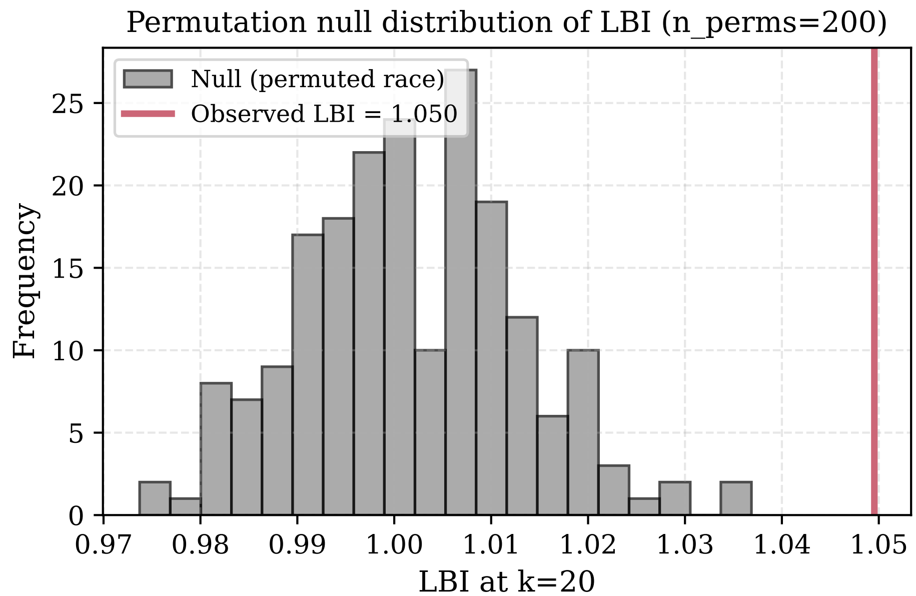

Figure 2: Permutation null distribution of LBI at neighbourhood scale $k = 20$, computed from 200 random race-label permutations. The null is centred at 1.002 with standard deviation 0.012. The observed LBI of 1.050 (red line) lies approximately 4 standard deviations above the null mean. None of the 200 permutations exceeded the observed value; the one-sided permutation p-value is below $1/200 = 0.005$.

1.038). Were those contested proxies manufacturing the effect, removing them would erase it; instead the signal persists, and including them raises LBI only from 1.038 to 1.050. (Adding `sex` as a further matching control raises LBI(race) slightly, to 1.060: matching within sex removes cross-sex dilution of the race contrast, sharpening the estimate rather than weakening it.) Second, the disparity does not depend on treating the raw decile as the unit of analysis—the objection that the legally salient decision is the *category*, not the exact score. Measured at the decision-relevant unit the effect is *larger*: LBI is 1.066 on the three-category (low/medium/high) mapping and 1.090 on the binary high-risk classification, i.e. matched cross-race pairs are about 9% more likely to straddle the high-risk threshold than matched same-race pairs.

These checks do not resolve the deeper omitted-variable concern—the COMPAS inputs absent from the public file (substance use, education, employment) cannot be tested here—so we continue to frame LBI as a conditional matched-comparison diagnostic (§6), not a proof of case-level unfairness. But the signal is stable across every control set the public data permit and strengthens at the unit that actually drives decisions.

5.7 Summary

Table 4 summarizes the empirical findings.

The case-level excess sensitivity to race is small in absolute magnitude (5 %) but is statistically robust: a 4-sigma effect above the permutation null, with a 95 % bootstrap CI that does not include 1, replicated across six neighbourhood scales spanning two orders of magnitude in k.

Table 4: Headline fairness measurements on the COMPAS Black/White subset, $n = 5{,}278$.

Metric	Value	Interpretation
Disparate impact (Black/White, high-risk)	1.741	Above the EEOC four-fifths threshold; large
FPR disparity (Black − White)	+0.203	ProPublica's headline finding; large
FNR disparity (Black − White)	−0.212	Complementary rate (distinct denominator); large
Predictive-parity gap, Pr(recid | high) (Black − White)	+0.055	Northpointe's defence; small
LBI(race) at $k = 20$	**1.050**	**95 % CI** $[1.040, 1.058]$**; permutation** $p < 0.005$

6 Discussion

6.1 What LBI adds to the fairness literature

LBI complements rather than replaces the existing fairness metrics. Three properties distinguish it:

(i) It is case-level, not population-level. The Chouldechova–Kleinberg impossibility is a constraint on population-level statistics. LBI evaluates the algorithm on matched pairs of individuals, so the impossibility does not constrain it. An algorithm can be case-level fair (LBI = 1) without satisfying either calibration or equalized odds; conversely, an algorithm can satisfy either calibration or equalized odds without being case-level fair.

(ii) It is a single number with a CI. Population-level metrics typically come in matched pairs (FPR and FNR; precision at high and low; positive-prediction rate by group). LBI is a scalar with a confidence interval, which is convenient for regulatory and audit reporting. Scalar simplicity is a reporting convenience, not an epistemic virtue: case-level fairness may itself vary by offence type, decision point, or assessment window, so a single index is best read as a screening signal to be disaggregated when it fires, not a substitute for the fuller metric panel.

(iii) It depends on a metric on legitimate predictors, not on the algorithm's internals. LBI can be computed by an auditor who has access to the inputs and outputs of the algorithm but not to its parameters. This is the typical audit setting: a regulator or oversight body can observe predictions but not the model.

6.2 Relationship to individual fairness

Dwork et al. (2012) introduced individual fairness as the requirement that "similar individuals receive similar predictions," formalized as a Lipschitz constraint with respect to a task-appropriate metric. They acknowledged the difficulty of specifying the metric and treated it as an open problem.

LBI is a quantitative implementation of individual fairness with two specific choices: the metric is the Mahalanobis distance on legitimate predictors, and the constraint is the matched-pair k-NN ratio rather than a uniform Lipschitz bound. The matched-pair formulation is operationally tractable in a way that a uniform Lipschitz bound is not—to verify a uniform Lipschitz bound, one must evaluate the algorithm on all pairs in the domain, which is infeasible for any non-trivial population.

6.3 Relationship to counterfactual fairness

Kusner et al. (2017) require that the algorithm's output be invariant under a counterfactual change of the protected attribute, holding the legitimate features fixed at their observed values. This is a strict version of case-level fairness; LBI is a soft version. The strict version requires a causal model of how the legitimate features depend on the protected attribute; the soft version (LBI) does not. The two are related: LBI $\to 1$ in the limit of perfect counterfactual fairness (a counterfactual flip of the protected attribute then moves the score no more than within-group variation in legitimate features does), and LBI meaningfully above 1 implies counterfactual unfairness if a counterfactual model is available.

6.4 Limitations

Choice of legitimate features, and contested proxies. The LBI signal depends on what we declare "legitimate," and that choice is normative, not merely practical—data availability cannot substitute for legal justification. Two failure modes cut in opposite directions. If relevant legitimate predictors are *omitted*, matched pairs are not truly similarly situated and LBI can *overstate* unfairness; the public ProPublica file omits COMPAS inputs such as substance-use, education, and employment items, so our matching is necessarily on a subset. Conversely, some included features are themselves *racially patterned proxies*—prior counts and juvenile counts reflect policing, surveillance, and charging practices as much as underlying conduct—so controlling on them can *understate* unfairness by normalizing part of the very disparity we seek to measure. These features are therefore not "uncontroversial"; a careful audit distinguishes predictors that are legitimate-and-available, legitimate-but-unavailable, and available-but-contested. The robustness analysis (§5.6) shows the case-level signal survives every available-feature choice we can test—including *dropping* the contested juvenile counts—but the unmeasured COMPAS inputs remain a genuine limit on any claim of complete case-level similarity.

Binary protected attributes. Definition 1 treats the protected attribute as binary. Multi-class extension is straightforward in either of two ways: compute pairwise LBI between all pairs of classes (giving a matrix of values), or take a one-vs-rest formulation. The COMPAS analysis uses the African-American vs. Caucasian binary because that is the contrast that drove the public controversy; the extension to multiple race categories is mechanical.

The 5 % magnitude. The COMPAS LBI of 1.050 is statistically robust but quantitatively modest. The legal and regulatory significance of a 5 % case-level excess sensitivity is a policy question, not a statistical one. Compare to a 100 % violation, which would correspond to an algorithm that flips its score entirely on the protected attribute—this is the calibration benchmark, not the threshold for legal concern.

Mahalanobis on $\mathbb{R}^d$. For continuous and binary features the Mahalanobis metric is well-defined; for genuinely categorical features (e.g., charge type with twenty levels), a modified metric (Gower distance, learned similarity, propensity score) may be more appropriate. The COMPAS feature space is predominantly continuous so this issue does not arise here.

Causal disentanglement. LBI measures a statistical association between the protected attribute and the algorithm's output, controlling for legitimate features via the Mahalanobis neighbourhood. It does not by itself establish a causal mechanism for that association. The causal story may be that the algorithm encodes race indirectly through correlated features that are not in our legitimate-feature set (residential zip code, employment history). Distinguishing these causal pathways is beyond what LBI alone can do; LBI is an audit signal that flags the issue and quantifies its magnitude, not a complete causal account.

7 Policy and Regulatory Implications

7.1 An audit tool for legal AI

The LBI can be operationalized as a regulatory audit signal for any algorithmic system that ingests legitimate predictors and a protected attribute and produces a score or classification. The procedure is:

1. Obtain a sample of the algorithm's inputs and outputs (n in the thousands; the COMPAS analysis used $n = 5{,}278$).
2. Specify the legitimate-feature set (this is the normative input that the audit cannot supply; it must come from statute, regulation, or expert designation).
3. Compute LBI(protected attribute) at multiple neighbourhood scales k.
4. Report the value, confidence interval, and permutation p-value.

The procedure does not require access to the algorithm's parameters, training data, or vendor documentation. It treats the algorithm as a black box and audits its outputs directly. This makes it well-suited to settings where the algorithm is proprietary, as COMPAS is.

7.2 Threshold setting

A regulatory threshold for LBI must come from policy, not statistics. Three descriptive anchors (offered for orientation, not as proposed thresholds):

LBI = 1.00 **to** 1.02**:** indistinguishable from no excess sensitivity at typical sample sizes. Probably acceptable.

LBI = 1.02 **to** 1.10**:** small but statistically detectable. The COMPAS result (1.050) sits in this range. Whether this is acceptable is a judgement call; the LBI value is one of several inputs (along with the disparate impact ratio, equalized-odds disparities, etc.) that a regulator should weigh.

LBI > 1.10**:** large excess sensitivity; the algorithm's outputs are more responsive to the protected attribute than to substantial legitimate-feature variation. Probably warrants mitigation.

We do not propose specific regulatory thresholds; we propose that LBI be added to the audit toolkit and that thresholds be set in consultation with domain expertise.

7.3 What LBI does not do

LBI is not a substitute for the constitutional and statutory analysis that determines whether an algorithm's deployment is lawful. A high LBI does not, by itself, establish a violation of any specific legal rule (the Equal Protection Clause, Title VII, the Fair Housing Act, etc.). A low LBI does not, by itself, license deployment. LBI provides a quantitative input to legal analysis; the legal analysis itself proceeds under the doctrinal categories appropriate to the deployment context.

8 Conclusion

The COMPAS controversy crystallized a structural ambiguity in algorithmic fairness: when group base rates differ, no scoring rule can simultaneously satisfy calibration and equalized error rates. The literature has responded by proliferating fairness criteria, each capturing a different facet, none of them dispositive.

We have introduced the Legal Bond Index, a measure that sits on a distinct axis. It does not constrain population-level statistics; it constrains case-level matched-pair behaviour. On the ProPublica COMPAS dataset, LBI(race) is 1.050 at neighbourhood scale $k = 20$ with 95 % CI $[1.040, 1.058]$ and permutation $p < 0.005$. The finding is small in magnitude but statistically robust, and it sits orthogonally to the calibration–equalized-odds debate. It is a quantitative implementation of Aristotle's like-cases-alike criterion; it is cheap to compute; it requires only the algorithm's inputs and outputs; and it produces a single number with a confidence interval suitable for regulatory and audit reporting.

The reference implementation is released as an open-source Jupyter notebook at `https://github.com/[author-handle]/lbi-compas` (repository URL pending acceptance; pre-release available on request). The data, the filter, the metric, and the figures are all reproducible from the public ProPublica file.

Author note. The geometric framework that motivates LBI (the Judicial Complex; the Legal Invariance Principle) is developed in companion theoretical work (Bond, 2026a,b). The present paper is self-contained: a reader interested only in LBI as an audit tool needs no part of that machinery.

Data availability. The COMPAS data are public at `https://github.com/propublica/compas-analysis`. The analysis notebook, the produced CSV tables, and the figures are bundled with the manuscript submission.

Competing interests. The author declares no competing interests.

Funding. No external funding supported this work.

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